

Adam White

Associate Professor · Canada CIFAR AI Chair · Fellow, Alberta Machine Intelligence Institute

Department of Computing Science, University of Alberta, Edmonton, AB, Canada T6G 2N8

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Research Interests

Reinforcement learning, continual learning, knowledge representations, representation learning, time series prediction, robotics, and machine learning.

Academic Positions

Associate Professor

Department of Computing Science, University of Alberta
Canada CIFAR AI Chair
Fellow of the Alberta Machine Intelligence Institute (Amii)
Reinforcement Learning and Artificial Intelligence Lab PI

2025–Present

2019–Present

2023–Present

2018–Present

Director of Scientific Operations

Alberta Machine Intelligence Institute (Amii)

2023–2026

Assistant Professor

Department of Computing Science, University of Alberta

2019–2025

Research Associate & Adjunct Professor

Department of Computing Science, University of Alberta

2017–2019

Assistant Research Professor

Department of Computer Science, Indiana University

2016–2017

Industry Positions

Co-founder & Chief Science Officer

RLCore Technologies

2023–Present

Staff Research Scientist

DeepMind

2021–2023

Senior Research Scientist

DeepMind

2017–2021

Education

Postdoctoral Fellow

Indiana University, Department of Computer Science

2015–2016

Doctor of Philosophy in Computing Science

University of Alberta

Advisor: Professor Richard Sutton

Thesis topic: Developing a predictive approach to knowledge

2015

Master of Science in Computing Science

University of Alberta

Advisor: Professor Richard Sutton

Thesis topic: A Standard System for Benchmarking in Reinforcement Learning

2006

Publications

A star (*) beside an author name indicates an HQP (co-)supervised by me (from 2019 onward), italics indicates a senior researcher, and bold marks my name. DM indicates a project completed at Google DeepMind. Acceptance rate noted as a percentage, if available.

Journal Articles

- [64] **Gradient Iterated Temporal-Difference Learning**, Vincent, T., Gerhardt, K., Tripathi, Y., Maraqtan, H., **White, A.**, *White, M.*, Peters, J., D'Eramo, C. The Reinforcement Learning Journal, Aug 2026. (34%)
- [63] **Forager: a lightweight testbed for continual learning with partial observability in RL**, *Tang, S., *Xiong, X., Hakhverdyan, A., Patterson, A., *Adkins, J., He, J., *Elelimy, E., *Panahi, P., *White, M.*, **White, A.** The Reinforcement Learning Journal, Aug 2026. (34%)
- [62] **Revisiting FTA: A Sparse One-to-Many Activation for Reinforcement Learning**, *Lazar, T., Vandergrift, M., *White, M.*, **White, A.** The Reinforcement Learning Journal, Aug 2026. (34%)
- [61] **Dynamics Models for Offline Hyperparameter Selection in Real-World RL**, *Coblin, J., *Wang, H., *White, M.*, **White, A.** The Reinforcement Learning Journal, Aug 2026. (34%)
- [60] **Endpoint Replay: Compressing the Recency Buffer in Deep Reinforcement Learning**, *Panahi, P., *Ashrafi, A., Du, H., *Patterson, A., *White, M.*, **White, A.** The Reinforcement Learning Journal, Aug 2026. (34%)
- [59] **Investigating the Utility of Mirror Descent in Off-policy Actor-Critic**, *Neumann, S., He, J., **White, A.**, *White, M.* The Reinforcement Learning Journal, Aug 2025. (38%)
- [58] **Modelling human exploration with light-weight meta reinforcement learning algorithms**, *Ferguson, T., *Fyshe, A.*, **White, A.** The Reinforcement Learning Journal, Aug 2025. (38%)
- [57] **Deep Reinforcement Learning with Gradient Eligibility Traces**, Elelimy, E., Daley, B., *Patterson, A., *Machado, M.*, **White, A.**, *Bowling, M.*, *White, M.* The Reinforcement Learning Journal, Aug 2025. (38%)
- [56] **Electrophysiological signatures of the effect of context on exploration: Greater attentional and learning signals when exploration is costly**, *Ferguson, T., *Fyshe, A.*, **White, A.** Brain Research, Jan 2025.
- [55] **Empirical design in reinforcement learning**, *Patterson, A., *Neumann, S., *White, M.*, **White, A.** Journal of Machine Learning Research, Oct 2024. (~17%)
- [54] **Goal-Space Planning with Subgoal Models**, Lo, C., *Roice, K., *Panahi, P. M., *Jordan, S., **White, A.**, Mihucz, G., Aminmansour, F., *White, M.* Journal of Machine Learning Research, Sept 2024. (~17%)
- [53] **The Cliff of Overcommitment with Policy Gradient Step Sizes**. *Jordan, S., *Neumann, S., Kostas, J. E., **White, A.**, *Thomas, P. S.* The Reinforcement Learning Journal, Aug 2024. (39%)
- [52] **Investigating the Interplay of Prioritized Replay and Generalization**. *Panahi, P. M., Patterson, A., *White, M.*, **White, A.** The Reinforcement Learning Journal, Aug 2024. (39%)
- [51] **Harnessing Discrete Representations for Continual Reinforcement Learning**. *Meyer, E. J., **White, A.**, *Machado, M. C.* The Reinforcement Learning Journal, Aug 2024. (39%)
- [50] **The Cross-environment Hyperparameter Setting Benchmark for Reinforcement Learning**. *Patterson, A., *Neumann, S., *Kumaraswamy, R., *White, M.*, **White, A.** The Reinforcement Learning Journal, Aug 2024. (39%)
- [49] **Investigating the Properties of Neural Network Representations in Reinforcement Learning**, *Wang, H., Miah, E., *White, M.*, *Machado, M. C.*, Abbas, Z., *Kumaraswamy, R., Liu, V., **White, A.** AI Journal, May 2024.
- [48] **Humans adopt different exploration strategies depending on the environment**, *Ferguson, T. D., *Fyshe, A.*, **White, A.**, *Krigolson, O. E.* Computational Brain & Behavior, 2023.
- [47] **GVFs in the Real World: Making Predictions Online for Water Treatment**, Janjua, M. K., Shah, H., *White, M.*, Miah, E., *Machado, M. C.*, **White, A.** Special issue on reinforcement learning for real life. Machine Learning, 2023.
- [46] **Agent-State Construction with Auxiliary Inputs**, *Tao, R. Y., **White, A.**, *Machado, M. C.* Transactions on Machine Learning Research, 2023.
- [45] **Reward-Respecting Subtasks for Model-Based Reinforcement Learning**, *Sutton, R. S.*, *Machado, M. C.*, Holland, Z., Szepesvari, D., Timbers, F., Tanner, B., **White, A.** The Journal of Artificial Intelligence (AIJ), 2023. DM

- [44] **Investigating Action Encodings in Recurrent Neural Networks in Reinforcement Learning.** *Schlegel, M. K., Tkachuk, V., **White, A.**, *White, M.* Transactions on Machine Learning Research, 2022.
- [43] **No More Pesky Hyperparameters: Offline Hyperparameter Tuning for RL.** *Wang, H., *Sakhadeo, A., **White, A.**, Bell, J., Liu, V., *Zhao, X., *Liu, P., Kozuno, T., *Fyshe, A.*, *White, M.* Transactions on Machine Learning Research, 2022.
- [42] **A Generalized Projected Bellman Error for Off-policy Value Estimation in Reinforcement Learning,** *Patterson, A., **White, A.**, *White, M.* Journal of Machine Learning Research, 2022. (~17%)
- [41] **From eye-blinks to state construction: diagnostic benchmarks for online representation learning,** *Rafiee, B., Abbas, Z., *Ghiassian, S., *Kumaraswamy, R., *Sutton, R. S.*, *Ludvig, E.*, **White, A.** Adaptive Behavior, 2022.
- [40] **General Value Function Networks,** *Schlegel, M. K., *Jacobsen, A., Zaheer, A., *Patterson, A., **White, A.**, *White, M.* Journal of Artificial Intelligence Research, 2020.
- [39] **Adapting Behavior via Intrinsic Reward: A Survey and Empirical Study,** *Linke, C., Ady, N. M., Degris, T., *White, M.*, **White, A.** Journal of Artificial Intelligence Research, 2020.
- [38] **Multi-timescale Nexting in a Reinforcement Learning Robot,** Modayil, J., **White, A.**, *Sutton, R. S.* Adaptive Behavior, 2014.
- [37] **The Reinforcement Learning Competitions,** *Whiteson, S.*, Tanner, B., **White, A.** AI Magazine, 2010.
- [36] **RL-Glue: Language-independent Software for Reinforcement-learning Experiments,** Tanner, B., **White, A.** The Journal of Machine Learning Research, 2009.

Refereed Conference Articles

- [35] **Position: Lifetime tuning is incompatible with continual reinforcement learning.** *Mesbahi, G., *Panahi, P., Mastikhina, O., *Tang, S., *White, M.*, **White, A.** International Conference on Machine Learning, Summer 2025. (19.7%)
- [34] **A Method for Evaluating Hyperparameter Sensitivity in Reinforcement Learning.** *Adkins, J., *Bowling, M.*, **White, A.** Advances in Neural Information Processing Systems, Sep 2024. (25.8%)
- [33] **Real-Time Recurrent Learning using Trace Units in Reinforcement Learning.** Elelimy, E., **White, A.**, *Bowling, M.*, *White, M.* Advances in Neural Information Processing Systems, Sep 2024. (25.8%)
- [32] **Position: Benchmarking is Limited in Reinforcement Learning Research.** *Jordan, S. M., **White, A.**, *Castro da Silva, B.*, *White, M.*, *Thomas, P. S.* International Conference on Machine Learning, May 2024. (26%)
- [31] **Position: Application-Driven Innovation in Machine Learning.** *Rolnick, D.*, *Aspuru-Guzik, A.*, *Beery, S.*, *Dilkina, B.*, *Donti, P. L.*, *Ghassemi, M.*, *Kerner, H.*, *Monteleoni, C.*, *Rolf, E.*, *Tambe, M.*, **White, A.** International Conference on Machine Learning, May 2024. (26%)
- [30] **Puzzle Entropy: Modeling Difficulty, Enjoyment and Learning.** *Chen, E. Y., **White, A.**, *Sturtevant, N.R.* Artificial Intelligence and Interactive Digital Entertainment, 2023. (Oral, shortlisted for best paper, 27.6% acceptance rate for orals)
- [29] **Loss of Plasticity in Continual Deep Reinforcement Learning.** Abbas, Z., Zhao, R., Modayil, J., **White, A.**, *Machado, M. C.* International Conference on Lifelong Learning Agents, 2023. (Oral). DM
- [28] **Measuring and Mitigating Interference in Reinforcement Learning.** Liu, V., *Wang, H., *Tao, R. Y., Javed, K., **White, A.**, *White, M.* International Conference on Lifelong Learning Agents, 2023.
- [27] **Auxiliary task discovery through generate-and-test.** *Rafiee, B., *Ghiassian, S., Jin, J., *Sutton, R. S.*, Luo, J., **White, A.** International Conference on Lifelong Learning Agents, 2023.
- [26] **Greedy Actor-Critic: A New Conditional Cross-Entropy Method for Policy Improvement.** *Neumann, S., Lim, S., Joseph, A. G., *Pan, Y., **White, A.**, *White, M.* International Conference on Learning Representations, 2023. (32.0%)
- [25] **The In-Sample Softmax for Offline Reinforcement Learning.** Xiao, C., *Wang, H., *Pan, Y., **White, A.**, *White, M.* International Conference on Learning Representations, 2023. (Top 25% of accepted papers; 32.0%)
- [24] **Learning Expected Emphatic Traces for Deep RL.** Jiang, R., Zhang, S., Chelu, V., **White, A.**, *van Hasselt, H.* AAAI Conference on Artificial Intelligence, 2022. (15%). DM
- [23] **Continual auxiliary task learning.** *McLeod, M., Lo, C., *Schlegel, M. K., *Jacobsen, A., *Kumaraswamy, R., *White, M.*, **White, A.** Advances in Neural Information Processing Systems, 34, 2021. (26%)
- [22] **Emphatic Algorithms for Deep Reinforcement Learning.** Jiang, R., Zahavy, T., Xu, Z., **White, A.**, Hessel, M., Blundell, C., *van Hasselt, H.* International Conference on Machine Learning (ICML), 2021. (21%). DM

- [21] **Gradient Temporal-Difference Learning with Regularized Corrections.** *Ghiassian, S., *Patterson, A., Garg, S., Gupta, D., **White, A.**, *White, M.* International Conference on Machine Learning (ICML), 2020. (21%)
- [20] **Improving Performance in Reinforcement Learning by Breaking Generalization in Neural Networks.** *Ghiassian, S., *Rafiee, B., Lo, Y. L., **White, A.** International Conference on Autonomous Agents and Multi-Agent Systems (AAMAS), 2020. (23%)
- [19] **Training Recurrent Neural Networks Online by Learning Explicit State Variables.** Nath, S., Liu, V., *Li, X., Chan, A., **White, A.**, *White, M.* International Conference on Learning Representations (ICLR), 2020. (26.5%)
- [18] **Planning with Expectation Models**, Wan, Y., Zaheer, A., *Sutton, R. S.*, **White, A.**, *White, M.* The International Joint Conference on Artificial Intelligence (IJCAI), 2019. (17.9%)
- [17] **Prediction in Intelligence: An Empirical Comparison of Off-policy Algorithms on Robots**, *Rafiee, B., *Ghiassian, S., **White, A.**, *Sutton, R. S.* The 18th International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2019. (24.3%)
- [16] **Meta-descent for Online, Continual Prediction**, *Jacobsen, A., *Schlegel, M. K., *Linke, C., Degris, T., **White, A.**, *White, M.* AAAI Conference on Artificial Intelligence (AAAI), 2019. (16.2%)
- [15] **Context-dependent Upper-confidence Bounds for Directed Exploration**, *Kumaraswamy, R., *Schlegel, M. K., **White, A.**, *White, M.* Advances in Neural Information Processing Systems (NeurIPS), 2018. (21.1%)
- [14] **Comparing Direct and Indirect Temporal-Difference Methods for Estimating the Variance of the Return**, Sherstan, C., Bennett, B., Young, K., Ashley, D., **White, A.**, *White, M.*, *Sutton, R. S.* Conference on Uncertainty in Artificial Intelligence (UAI), 2018. (30.8%)
- [13] **Organizing Experience: a Deeper Look at Replay Mechanisms for Sample-based Planning in Continuous-State Domains**, Pan, Y., Zaheer, A., **White, A.**, *Patterson, A.*, *White, M.* International Joint Conference on Artificial Intelligence (IJCAI), 2018. (20.5%)
- [12] **Accelerated Gradient Temporal Difference Learning**, Pan, Y., **White, A.**, *White, M.* AAAI Conference on Artificial Intelligence, 2017. (24.6%)
- [11] **Introspective Agents: Confidence Measures for General Value Functions**, Sherstan, C., Machado, M.C., **White, A.**, *Pilarski, P. M.* Artificial General Intelligence (AGI), 2016.
- [10] **Investigating Practical Linear Temporal Difference Learning**, **White, A.**, *White, M.* International Conference on Autonomous Agents and MultiAgent Systems (AAMAS), 2016. (24.9%)
- [9] **A Greedy Approach to Adapting the Trace Parameter for Temporal Difference Learning**, *White, M.*, **White, A.** International Conference on Autonomous Agents and MultiAgent Systems (AAMAS), 2016. (24.9%)
- [8] **Scaling Life-long Off-policy Learning**, **White, A.**, Modayil, J., *Sutton, R. S.* IEEE International Conference on Development and Learning and Epigenetic Robotics (ICDL), 2012. (Distinguished Paper Award; 65%)
- [7] **Multi-timescale Nexting in a Reinforcement Learning Robot**, Modayil, J., **White, A.**, *Sutton, R. S.* From Animals to Animats, 2012.
- [6] **Acquiring Diverse Predictive Knowledge in Real Time by Temporal-difference Learning**, Modayil, J., **White, A.**, *Pilarski, P. M.*, *Sutton, R. S.* Systems, Man, and Cybernetics, 2012.
- [5] **Horde: A Scalable Real-time Architecture for Learning Knowledge from Unsupervised Sensorimotor Interaction.** *Sutton, R. S.*, Modayil, J., Delp, M., Degris, T., *Pilarski, P. M.*, **White, A.**, *Precup, D.* International Conference on Autonomous Agents and Multiagent Systems, 2011. (21.9%)
- [4] **Interval Estimation for Reinforcement-Learning Algorithms in Continuous-State Domains**, *White, M.*, **White, A.** Advances in Neural Information Processing Systems (NIPS), 2010. (24.04%)
- [3] **Feature Construction for Reinforcement Learning in Hearts**, Sturtevant, N., **White, A.** Computers and Games, 2007.

Thesis

- [2] **A Standard System for Benchmarking in Reinforcement Learning**, *White, A.*, Master's Thesis, University of Alberta, 2006.
- [1] **Developing a predictive approach to knowledge**, *White, A.*, Doctoral Thesis, University of Alberta, 2015.

Research Grants

NSERC Discovery Grant	<i>2026–2031</i>
\$300,000. Sole PI: “Continual Reinforcement Learning for the Real World”.	
Canada CIFAR Catalyst Grant	<i>2025–2027</i>
\$50,000 over two years. PI: “Adapting lighting to improve plant growth via reinforcement learning”.	
Canada CIFAR AI Chair Renewal	<i>2025–2030</i>
\$500,000 over five years. Sole PI.	
Canada CIFAR AI Recruitment Chair	<i>2020–2025</i>
\$405,000 over five years. Sole PI: “Intrinsically Motivated, Life-long RL”.	
NSERC Discovery Grant	<i>2019–2026</i>
\$196,000 plus a \$12,500 launch supplement for the first year. Sole PI: “Developing inquisitive, model-based agents for reinforcement learning”.	
NSERC CRD	<i>2018–2022</i>
\$536,700 total for three years. Co-PI: “Optimizing water treatment operation using reinforcement learning.”	

Supervision

Current Students

Baxter Madore, MSc	<i>2025–present</i>
Marie Del Valle, MSc	<i>2025–present</i>
Nathan Zeweniuk, MSc	<i>2025–present</i>
Steven Tang, MSc	<i>2024–present</i>
Armin Ashrafi, MSc	<i>2024–present</i>
Oliver Diamond, MSc	<i>2024–present</i>
Sam Scholnick-Hughes, MSc	<i>2024–present</i>
Eric Xiong, MSc	<i>2024–present</i>
Jacob Adkins, PhD (co-advised with Bowling)	<i>2025–present</i>
Parham Mohammad Panahi, PhD (co with Bowling)	<i>2026–present</i>
Samuel Neumann, PhD	<i>2022–present</i>
Blanca Miller (Doctoral Supervisory Committee)	<i>2024–present</i>
Saeed Najafi (Doctoral Supervisory Committee)	<i>2023–present</i>
Alex Lewandowski (Doctoral Supervisory Committee)	<i>2022–present</i>

Below *co-advised indicates I was the primary advisor, otherwise the supervision was equal.

Graduated

Anffany Chen, Post-doc	<i>2026</i>
Cameron Jen, MSc	<i>2026</i>
Montaser Mohammedalamen, Doctoral Supervisory Committee	<i>2025</i>
Ty Lazar, MSc	<i>2025</i>
Tom Ferguson, Post-doc (co-advised with Fyshe)	<i>2025</i>
Parham Mohammad Panahi, RA	<i>2025</i>
Hongming Zhang, Doctoral Supervisory Committee	<i>2025</i>
Han Wang (co-advised with White), PhD	<i>2025</i>
Jacob Adkins (co-advised with Bowling), MSc	<i>2025</i>
Scott Jordan (co-advised with White), Post-doc	<i>2024</i>
Ehsan Imani, Doctoral Supervisory Committee	<i>2024</i>
Andrew Jacobsen, Doctoral Supervisory Committee	<i>2024</i>
Golnaz Mesbahi (*co-advised with White), MSc	<i>2024</i>
Parham Mohammad Panahi, MSc	<i>2024</i>
Kevin Roice (co-advised with White), MSc	<i>2024</i>
Jordan Coblin, MSc	<i>2024</i>
Banafsheh Rafiee (co-advised with Sutton), PhD	<i>2024</i>
Andrew Patterson, Doctoral Supervisory Committee	<i>2023</i>
Matthew Schlegel (co-advised with White), PhD	<i>2023</i>

Edan Meyer (co-advised with Machado), MSc	2023
Subhojeet Pramanik (co-advised with Machado), MSc	2023
Eugene Chen (co-advised with Sturtevant), MSc	2023
Kailash Seshadri, Summer Ugrad	2023
Cameron Jen, NSERC USRA	2023
Derek Li, MSc	2022
Paul Liu, MSc	2022
Samuel Neumann, MSc	2022
David Tao (co-advised with Machado), MSc	2022
Sina Ghiassian (co-advised with Sutton), PhD	2022
Raksha Kumaraswamy (co-advised with White), PhD	2021
Kris De Asis, Doctoral Supervisory Committee	2021
Matthew McLeod, MSc	2021
Archit Sakhadeo (*co-advised with Fyshe), MSc	2021
Xutong Zhao, MSc	2021
Yangchen Pan, Doctoral Supervisory Committee	2021
Cam Linke (*co-advised with Sutton), MSc	2020
Han Wang, MSc	2020
Niko Yasui (co-advised with White), MSc	2020
Andrew Jacobsen, MSc	2019
Banafsheh Rafiee, Research Associate	2019
Banafsheh Rafiee (co-advised with Sutton), MSc	2018
Andrew Jacobsen, NSERC USRA	2018

Thesis Examining Committee

Dylan Miller, University of Alberta, MSc Examiner	2026
Christoph Bumiller, University of Bath (UK), PhD External Examiner	2026
Amirhossein Rajabpour, University of Alberta, MSc Examiner	2025
Mahshid Rahmani Hanzaki, University of Alberta, MSc Examiner	2025
Videh Raj Nema, University of Alberta, MSc Examiner	2025
Geraud Nangue Tasse, Witwatersrand, PhD External	2024
Tales Henrique Carvalho, University of Alberta, MSc Examiner	2023
Muhammad Kamran Janjua, University of Alberta, MSc Examiner	2023
Chenjun Xiao, University of Alberta, PhD Examiner	2022
Calarina Muslimani, University of Alberta, MSc Examiner	2022
Chunlok Lo, University of Alberta, MSc Examiner	2022
Gabor Mihucz, University of Alberta, MSc Examiner	2022
Ethan Jones, UNSW Sydney, PhD External Examiner	2022
Yannis Flet-Berliac, Inria Lille, PhD External Examiner	2021
Joshua Romoff, McGill, PhD External Examiner	2021
Chen Ma, University of Alberta, MSc final	2019
Arta Seify, University of Alberta, MSc final	2019

Teaching

Reinforcement Learning Four Course Specialization

Fall 2019–Present

Coursera and University of Alberta (undergraduate level). >100,000 registrations

INT-D 161: AI Everywhere

Fall 2025, Winter 2025, Winter 2024

University of Alberta (undergraduate)

CMPUT 655: Reinforcement Learning I

Fall 2022

University of Alberta (graduate)

CMPUT 365: Introduction to Reinforcement Learning

Fall 2021

University of Alberta (undergraduate)

CMPUT 607: Empirical Reinforcement Learning

Winter 2021

University of Alberta (graduate)

CMPUT 397: Reinforcement Learning I

Fall 2019

University of Alberta (undergraduate)

CMPUT 366: Intelligent Systems

Fall 2017, 2018

University of Alberta (undergraduate)

CMPUT 609: Reinforcement Learning for Artificial Intelligence

Fall 2017

University of Alberta (graduate)

CS B659: Reinforcement Learning

Spring 2016, 2017

Indiana University (graduate)

Talks

- Invited talk, Amii UpperBound Conference. *Reinforcement Learning for the Real World*, May 2026.
- Invited talk, CIFAR workshop on Leveraging Insights from Infant Learning to Advance Machine Learning. *Continual Open-ended Reinforcement Learning*, March 2026.
- Invited Lecture, CIFAR Deep Learning and Reinforcement Learning Summer School. *Empirical AI Research*, Summer 2025.
- Invited Lecture, AI4Good Lab. *Introduction to Reinforcement Learning: Control*, May 2025.
- Invited Lecture, AI4Good Lab. *Introduction to Reinforcement Learning: Prediction*, May 2025.
- Invited Lecture, Deep RL Course Sharif University of Technology. *Reinforcement Learning Experiments That Matter!*, May 2025.
- Invited talk, Amii UpperBound Conference. *Investigating the Properties of Neural Network Representations in Reinforcement Learning*, May 2025.
- Invited talk, Autonomous Agents Research Group, University of Edinburgh (virtual). *Reinforcement Learning Experiments That Matter!*, January 2025.
- Invited talk, Edmonton Public Schools Student AI Conference. *AI Past, Present, and Future*, Spring 2024.
- Invited talk, Vector Seminar, Vector Institute, Toronto. *Reinforcement Learning Experiments That Matter!*, June 2024.
- Invited talk, Bayes-Duality Workshop, Riken Institute, Japan. *Continual Subtask Learning*, June 2024.
- Invited Lecture, City of Edmonton: Lunch and Learn Series. *AI at UofA*, June 2024.
- Invited talk, Edmonton Data Management Association Meeting. *AI Past, Present, and Future*, June 2024.
- Invited Lecture, AI4Good Lab. *Introduction to Reinforcement Learning: Control*, May 2024.
- Invited Lecture, AI4Good Lab. *Introduction to Reinforcement Learning: Prediction*, May 2024.
- University of Alberta, Computing Science Department Seminar. *Continual Reinforcement Learning in the Real World*, May 2024.
- Invited talk, Amii UpperBound Conference. *Continual Subtask Learning*, May 2024.
- Invited talk, Amii UpperBound Conference. *Better RL Experiments*, May 2024.
- Invited talk, Edmonton Public Library: On the Edge. *AI Past, Present, and Future*, Spring 2024.
- Invited Journal Track Talk, AAAI Conference on Artificial Intelligence. *Reward-respecting Subtasks for Model-based Reinforcement Learning*, Spring 2024.
- Invited talk, Stanford University RL Forum. *Continual Subtask Learning*, Fall 2023.
- Invited talk, Edmonton Regional Innovation Network, Northern Alberta Institute of Technology. *The future of automated decision making*, Fall 2023.
- Invited talk, Pan-Canadian AI Strategy Symposium on AI for Energy and the Environment. *AI for safe and sustainable rural drinking water*, Spring 2023.
- Invited talk, University of Bath (virtual). *Leveraging offline data for learning in deployment*, Spring 2023.
- Invited talk, RL-sofa Seminar Series, Mila Montreal (virtual). *Leveraging offline data for learning in deployment*, Spring 2023.
- Invited lecture, MIT Deep Reinforcement Learning Class (virtual). *Better Experiments in Reinforcement Learning*, Fall 2022.
- Invited Lecture, CIFAR Deep Learning and Reinforcement Learning Summer School (virtual). *Better Experiments in Reinforcement Learning*, Summer 2022.
- Invited talk, Amii AI Week Academic Symposium, Edmonton Alberta. *Improving Water Treatment with Reinforcement*

Learning, Spring 2022.

- Invited Lecture, CIFAR Deep Learning and Reinforcement Learning Summer School, Montreal Quebec (virtual). *Introduction to Reinforcement Learning*, Summer 2020.
- Keynote Lecture, Inaugural Autonomous Systems Initiative (ASI) Workshop. *Scalable Predictive Knowledge Acquisition*, May 2020.
- Invited talk, Huawei STW Summit, Shenzhen China. *Parallel Prediction learning with Reinforcement Learning*, May 2019.
- Invited Lecture, CIFAR Deep Learning and Reinforcement Learning Summer School, Edmonton Alberta. *Introduction to Reinforcement Learning*, Summer 2019.
- Invited talk, Workshop on Deep, fast and shallow learning in humans and machines, Indiana University. *Curiosity for learning about many things in parallel*, May 2018.
- Invited talk, The 11th Barbados Workshop on Reinforcement Learning. *Realtime Interactive Mastery*, March 2018.
- Invited talk, Indiana University Department of Statistics Colloquium Series. *Continual prediction learning on robots*, November 2016.
- Invited talk, International Conference on Autonomous Agents and Multiagent Systems, Singapore. *Investigating practical linear temporal difference learning*, May 2016.
- Invited talk, Intelligent and Interactive Systems Seminar, Indiana University. *Developing a predictive approach to knowledge*, April 2016.
- Invited talk, The 9th Barbados Workshop on Reinforcement Learning. *Experiences trying to put it all together*, April 2015.
- Invited talk, AAAI Workshop on Decision Making With Big Data. *Surprise and curiosity in big data reinforcement learning*, July 2014.
- Invited talk, International Conference on Development and Learning and Epigenetic Robotics. *Scaling life-long off-policy learning*, December 2012.
- Invited talk, Flowers Group, INRIA France. *Parallel off-policy knowledge acquisition*, September 2012.
- Invited talk, International Workshop on Evolutionary and Reinforcement Learning for Autonomous Robot Systems. *Acquiring a Broad Range of Empirical Knowledge in Real Time by Temporal-Difference Learning*, August 2012.
- Invited talk, The 7th Barbados Workshop on Reinforcement Learning. *Real-time Off-policy Learning from Big Data in Robotics*, April 2012.
- Invited talk, The 6th Barbados Workshop on Reinforcement Learning. *Multi-time-scale Nexting on a Reinforcement Learning Robot*, April 2011.
- Invited talk, International Conference on Autonomous Agents and Multiagent Systems, Taipei. *Horde: A Scalable Real-time Architecture for Learning Knowledge from Unsupervised Sensorimotor Interaction*, May 2011.
- Invited talk, The 5th Barbados Workshop on Reinforcement Learning. *Feature Selection for Hearts and a Trail Towards Feature Discovery for a Mobile Robot*, March 2010.
- Invited talk, Advances in Neural Information Processing Systems, Reinforcement Learning Workshop: Benchmarks and Bakeoffs II. *A New Evaluation Framework for Reinforcement Learning Experiments*, December 2005.

Service and Outreach

Editorial and Review Service

- Editor in Chief, The Reinforcement Learning Journal (RLJ)
- Action Editor, The Journal of Machine Learning Research (JMLR)
- Action Editor, Transactions on Machine Learning Research (TMLR)
- Innovation, Science and Economic Development Canada Safe and Secure AI Advisory Group
- Killam Conversation: The Next 50 Years of Artificial Intelligence, Massey College
- Innovation Panel, University of Alberta Board of Governors
- Canada-Italy Joint Advisory Group on AI (rep for Government of Canada)
- Senior Area Chair, International Conference on Learning Representations
- ICLR 2021 Outstanding Paper Award Committee 2021
- Senior Program Committee, AAAI Conference on Artificial Intelligence
- Area Chair, Neural Information Processing Systems

- Area Chair, International Conference on Learning Representations
- Area Chair, International Conference on Machine Learning
- Program Committee, International Conference on Machine Learning
- Program Committee, International Joint Conference on Artificial Intelligence
- Program Committee, Artificial Intelligence and Statistics Conference
- Reviewer, Conference on Autonomous Agents and Multi-agent Systems
- Reviewer, Transactions on Computational Intelligence and AI in Games
- Reviewer, European Workshop on Reinforcement learning
- Reviewer, Journal of Machine Learning Research
- Reviewer, Artificial Intelligence Journal
- Reviewer, Journal of Robotics
- Reviewer, Journal Computational Intelligence and AI in Games

Event Organization

- Tutorial Co-chair, International Conference on Machine Learning, 2026
- Review-process Chair, the Reinforcement Learning Conference, 2026
- Organizer, NSERC G7+ AI Expert Summit (AI for Science), University of Ottawa, 2025
- General Chair, The Reinforcement Learning Conference, 2025
- Program Chair, The Reinforcement Learning Conference, 2024
- Board Member, Conference on Lifelong Learning Agents, 2023–2025
- Amii UpperBound (formerly AI-Week), *Organizing Committee*, 2023, 2024
- CIFAR Pan-Canadian AI Strategy AI for Energy and the Environment, *Working Group*, 2023
- CIFAR Deep Learning and Reinforcement Learning Summer School, *Organising Committee*, 2019, 2021, 2022, 2025
- Founding member of *mementor.net*: a platform connecting AI researchers across the world
- A Roadmap to Never-Ending RL, International Conference on Learning Representations, *Main Organiser*, 2019
- International Conference on Machine Learning, *Social Chair*, 2021
- International Conference on Machine Learning, *Social Chair*, 2020
- International Conference on Learning Representations, *Social Chair*, 2020
- Multi-disciplinary Conference on Reinforcement Learning and Decision Making workshop: Curiosity for Decision Making, *Main Organiser*, 2019
- International Conference on Machine Learning, Reinforcement Learning Competition, *Organising Committee*, 2008
- Advances in Neural Information Processing Systems workshop: The First Annual Reinforcement Learning Competition, *Main Organiser*, 2007
- Advances in Neural Information Processing Systems workshop: Reinforcement Learning Benchmarks and Bakeoffs II, *Organising Committee*, 2006

Awards

- *Canada CIFAR AI Chair Renewal*, 2025, CIFAR Canada
- *Highlighted Area Chair*, 2022 International Conference on Learning Representations
- *Canada CIFAR AI Chair*, 2020, CIFAR Canada
- *Top 200 Reviewer Award (highest award)*, 2019 International Conference on Neural Information Processing Systems
- *Top 10 Reviewer Award*, 2018 International Conference on Machine Learning
- *Best Reviewer Award*, 2017 International Conference on Neural Information Processing Systems
- *Reviewer Award*, 2015 International Conference on Machine Learning
- *Paper of Distinction*, IEEE International Conference on Developmental Robotics and Epigenetic Robotics, 2012
- *Best Paper*, International Workshop on Evolutionary and Reinforcement Learning for Autonomous Robot Systems, 2012

Code Released

RL-Glue version 1.0*2006*

LANGUAGES: JAVA, C, AND PYTHON. A language-independent communication protocol and evaluation framework for reinforcement learning experiments.

RL-Glue version 2.0*2007*

LANGUAGES: JAVA, C, AND PYTHON. A version of RL-Glue that allowed benchmarking competitions to be run via remote socket communication.

Personal Information

Citizenship: Canada

Languages: English